

SOLIDWORKS API (.NET VBA C#) — Reference (201/337): IModelDocExtension (part 7 of 8)

SOLIDWORKS API Documentation (converted from CHM)

IModelDocExtension (part 7 of 8) (Type)

SOLIDWORKS API Help

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PrintOut4 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PrintOut4 Method (IModelDocExtension)

Prints this document without displaying any dialogs or message boxes.

.NET Syntax

Visual Basic (Declaration)

```
Sub PrintOut4( _  
    ByVal Printer As System.String, _  
    ByVal PrintFileName As System.String, _  
    ByVal PrintSpecification As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Printer As System.String  
Dim PrintFileName As System.String  
Dim PrintSpecification As System.Object  
  
instance.PrintOut4(Printer, PrintFileName, PrintSpecification)
```

C#

```
void PrintOut4(  
    System.string Printer,  
    System.string PrintFileName,  
    System.object PrintSpecification  
)
```

C++/CLI

```
void PrintOut4(  
& System.String^ Printer,  
& System.String^ PrintFileName,  
& System.Object^ PrintSpecification  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Printer
Name of the printer to which to print (see **Remarks**)
PrintFileName
Name of file to which to print (see **Remarks**)
PrintSpecification
[IPrintSpecification](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PrintOut4](#).

Example

[Print Drawing \(VBA\)](#)
[Print Drawing \(VB.NET\)](#)
[Print Drawing \(C#\)](#)

Remarks

This method supports printing parts, assemblies, and both single and multisheet drawings.
Before calling this method:

1. Call [IModelDocExtension::GetPrintSpecification](#) to get the IPrintSpecification object for this document.
2. Set [IPrintSpecification::PrintToFile](#) to:
 - true to print to PrintFileName.
 - false to print to Printer.
3. Set other properties on the IPrintSpecification object.
4. Use the IPrintSpecification object to specify PrintSpecification.

If Printer, PrintFileName, and PrintSpecification are not specified, then this method prints to the default printer specified by [IModelDoc2::Printer](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::PrintDirect Method \(\)](#)
[IModelDoc2::PrintPreview Method \(\)](#)
[IModelDoc2::ClosePrintPreview Method \(\)](#)
[IModelDocExtension::SaveAs Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PublishSTEP242File Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PublishSTEP242File Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::PublishSTEP242File2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function PublishSTEP242File( _  
    ByVal Path As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Path As System.String  
Dim value As System.Integer  
  
value = instance.PublishSTEP242File(Path)
```

C#

```
System.int PublishSTEP242File(  
    System.string Path  
)
```

C++/CLI

```
System.int PublishSTEP242File(  
& System.String^ Path  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Path

Full qualified path to which to export the SOLIDWORKS MBD 3D part or assembly; use **.STP** for the file name extension

Return Value

Status as defined in [swStep242Error_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PublishSTEP242File](#).

Remarks

This method requires that the SOLIDWORKS MBD add-in be loaded. ([ISldWorks::LoadAddIn](#))

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::PublishTo3DPDF Method \(\)](#)
[IMBD3DPdfData::CreateAttachSTEP242 Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PublishSTEP242File2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PublishSTEP242File2 Method (IModelDocExtension)

Exports the SOLIDWORKS MBD 3D part or assembly to a STEP 242 file.

.NET Syntax

Visual Basic (Declaration)

```
Function PublishSTEP242File2( _  
    ByVal MBDSTEP242Data As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim MBDSTEP242Data As System.Object  
Dim value As System.Integer  
  
value = instance.PublishSTEP242File2(MBDSTEP242Data)
```

C#

```
System.int PublishSTEP242File2(  
    System.object MBDSTEP242Data  
)
```

C++/CLI

```
System.int PublishSTEP242File2(  
& System.Object^ MBDSTEP242Data  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[MBDSTEP242Data](#)
[IMBDSTEP242Data](#)

Return Value

Status as defined by [swStep242Error_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PublishSTEP242File2](#).

Example

See the IMBDSTEP242Data example.

Remarks

This method requires that the SOLIDWORKS MBD add-in be loaded. ([ISldWorks::LoadAddIn](#))

See the SOLIDWORKS Help for details about MBD.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::PublishTo3DPDF Method \(\)](#)
[IMBD3DPdfData::CreateAttachSTEP242 Property \(\)](#)

Availability

SOLIDWORKS 2025 SP03, Revision Number 33.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

PublishTo3DPDF Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PublishTo3DPDF Method (IModelDocExtension)

Creates a 3D PDF for SOLIDWORKS MBD.

.NET Syntax

Visual Basic (Declaration)

```
Function PublishTo3DPDF( _  
    ByVal MBDPdfData As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim MBDPdfData As System.Object  
Dim value As System.Integer  
  
value = instance.PublishTo3DPDF(MBDPdfData)
```

C#

```
System.int PublishTo3DPDF(  
    System.object MBDPdfData  
)
```

C++/CLI

```
System.int PublishTo3DPDF(  
& System.Object^ MBDPdfData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MBDPdfData
[SOLIDWORKS MBD 3D PDF data](#)

Return Value

Status as defined in [swPublishTo3DPDFError_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PublishTo3DPDF](#).

Example

[Publish Text and Custom Properties from Theme to MBD 3D PDF \(C#\)](#)
[Publish Text and Custom Properties from Theme to MBD 3D PDF \(VB.NET\)](#)
[Publish Text and Custom Properties from Theme to MBD 3D PDF \(VBA\)](#)
[Fire Notification When Publishing Part to MBD 3D PDF \(C#\)](#)

Remarks

SOLIDWORKS Model Based Definition (MBD) is an integrated drawingless manufacturing solution for SOLIDWORKS.

This method:

- requires that the SOLIDWORKS MBD add-in be loaded. ([ISldWorks::LoadAddin](#))
- corresponds to the **Publish to 3D PDF** button on the SOLIDWORKS MBD toolbar.

Typical steps for generating a SOLIDWORKS MBD 3D PDF using the SOLIDWORKS API are:

1. Get the MBD3DPdfData object using [IModelDocExtension::GetMBD3DPdfData](#).
2. Set the path and file name of the SOLIDWORKS MBD 3D PDF theme using [IMBD3DPdfData::ThemeName](#).
3. Get standard and custom views using [IMBD3DPdfData::GetStandardViews](#) and [IMBD3DPdfData::GetMoreViews](#).
4. Set standard and custom views using [IMBD3DPdfData::SetStandardViews](#) and [IMBD3DPdfData::SetMoreViews](#).
5. Get and set text and custom properties in the theme using [IMBD3DPdfData::GetTextAndCustomProperties](#) and [ITextAndCustomProperty](#).
6. Set the path for the SOLIDWORKS MBD 3D PDF using [IMBD3DPdfData::FilePath](#).
7. Generate the SOLIDWORKS MBD 3D PDF using this method.

See the SOLIDWORKS Help for details about MBD.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IView3D Interface](#)
[IModelDocExtension::PublishSTEP242File Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PurgeDisplayState Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PurgeDisplayState Method (IModelDocExtension)

Purges identical display states so that only unique display states remain.

.NET Syntax

Visual Basic (Declaration)

```
Function PurgeDisplayState() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
value = instance.PurgeDisplayState()
```

C#

```
System.bool PurgeDisplayState()
```

C++/CLI

```
System.bool PurgeDisplayState();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if identical display states are purged, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PurgeDisplayState](#).

Example

[Create, Unlink, and Purge Display States in Parts \(VB.NET\)](#)
[Create, Unlink, and Purge Display States in Part \(VBA\)](#)
[Create, Unlink, and Purge Display States in Parts \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IConfiguration::ApplyDisplayState Method](#)
[IConfiguration::CopyDisplayStateFromConfiguration Method](#)
[IConfiguration::CreateDisplayState Method](#)
[IConfiguration::DeleteDisplayState Method](#)
[IConfiguration::GetDisplayStates Method](#)
[IConfiguration::GetDisplayStatesCount Method](#)
[IModelDocExtension::LinkedDisplayState Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RayIntersections Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RayIntersections Method (IModelDocExtension)

Finds the intersections between the specified set of rays and the specified set of bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function RayIntersections( _  
    ByVal BodiesIn As System.Object, _  
    ByVal BasePointsIn As System.Object, _  
    ByVal VectorsIn As System.Object, _  
    ByVal Options As System.Integer, _  
    ByVal HitRadius As System.Double, _  
    ByVal Offset As System.Double, _  
    ByVal HighPrecision As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim BodiesIn As System.Object  
Dim BasePointsIn As System.Object  
Dim VectorsIn As System.Object  
Dim Options As System.Integer  
Dim HitRadius As System.Double  
Dim Offset As System.Double  
Dim HighPrecision As System.Boolean  
Dim value As System.Integer
```

```
value = instance.RayIntersections(BodiesIn, BasePointsIn, VectorsIn, Options, HitRadius, Offset, HighPrecision)
```

C#

```
System.int RayIntersections(  
    System.object BodiesIn,  
    System.object BasePointsIn,  
    System.object VectorsIn,  
    System.int Options,  
    System.double HitRadius,  
    System.double Offset,  
    System.bool HighPrecision  
)
```

C++/CLI

```

System.int RayIntersections(
& System.Object^ BodiesIn,
& System.Object^ BasePointsIn,
& System.Object^ VectorsIn,
& System.int Options,
& System.double HitRadius,
& System.double Offset,
& System.bool HighPrecision
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodiesIn

Array of [bodies](#) that are hit by the rays (see **Remarks**)

BasePointsIn

Array of doubles containing the x,y,z base points of each ray

VectorsIn

Array of doubles containing the direction vectors of each ray

Options

Options as defined in [swRayPtsOpts_e](#) (see **Remarks**)

HitRadius

Radius of hit cylinder; this is used mainly in grazing cases to establish a hit (see **Remarks**)

Offset

Length tolerance to use to establish whether a hit on a face represents the entry or exit of the ray from the body (see **Remarks**)

HighPrecision

True to use maximum precision when evaluating intersections close to the ray boundary, false to use dynamic precision based on the ray radius (see **Remarks**)

Return Value

Number of intersections

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RayIntersections](#).

Remarks

The method performs intersection operations between the specified bodies and the specified ray vectors. To get the results (a set of intersection points, intersection normals, and the bodies from your BodiesIn array that were hit), you must call [IModelDoc2::GetRayIntersectionsPoints](#) and [IModelDoc2::GetRayIntersectionTopology](#).

Information returned by IModelDoc2::GetRayIntersectionsPoints and IModelDoc2::GetRayIntersectionTopology depends partially on the value of the Options parameter. Valid Option values can be combined using bitwise operations. See IModelDoc2::GetRayIntersectionsPoints to determine which data is always output regardless of the values specified by Options.

For each ray that hits an edge or a vertex, the Offset value is added in both the positive and negative directions along the ray, and the points computed are tested for spatial inclusion in the hit body. This determines whether the ray was entering, leaving, or just grazing the body at the hit point. Entry and exit for faces can be computed and does not require an offset.

The difference between this method and the now obsolete IModelDoc2::RayIntersections and IModelDoc2::IRayIntersections is the HighPrecision parameter. Setting HighPrecision to true forces use of the highest possible precision when evaluating intersections close to ray profile boundaries. Setting HighPrecision to false forces use of a dynamic tolerance that is based on the HitRadius, which is the behavior of the obsolete methods.

The actual ray profile is polygonal, approximating an ideal circular ray profile of HitRadius. Evaluating intersections very close to the ray profile boundary may produce unexpected results due to this polygonal approximation. Experimentation with HighPrecision is recommended.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDoc2::MultiSelectByRay Method \(\)](#)

[IModelDoc2::SelectByRay Method \(\)](#)

Availability

SOLIDWORKS 2018 SP02, Revision Number 26.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

Rebuild Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Rebuild Method (IModelDocExtension)

Rebuilds the model in assembly and drawing documents and returns the status of the rebuild.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function Rebuild( _  
    ByVal Options As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Options As System.Integer  
Dim value As System.Boolean  
  
value = instance.Rebuild(Options)
```

C#

```
System.bool Rebuild(  
    System.int Options  
)
```

C++/CLI

```
System.bool Rebuild(  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

Type of rebuild as defined in [swRebuildOptions_e](#)

Return Value

True if the rebuild is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Rebuild](#).

Example

[Rebuild an Assembly \(VBA\)](#)
[Rebuild an Assembly \(VB.NET\)](#)
[Rebuild an Assembly \(C#\)](#)

Remarks

Use [IModelDoc2::GetType](#) to check the type of document.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::EditRebuild3 Method \(\)](#)
[IModelDoc2::ForceRebuild3 Method \(\)](#)
[IModelDocExtension::NeedsRebuild2 Property \(\)](#)
[IModelDocExtension::EditRebuildAll Method \(\)](#)
[IModelDocExtension::ForceRebuildAll Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

Updates the [3D Views](#) of this part or assembly.

.NET Syntax

Visual Basic (Declaration)

```
Sub Refresh3DViews()
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension
```

```
instance.Refresh3DViews()
```

C#

```
void Refresh3DViews()
```

C++/CLI

```
void Refresh3DViews();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Refresh3DViews](#).

Example

[Capture 3D View \(VBA\)](#)
[Capture 3D View \(VB.NET\)](#)
[Capture 3D View \(C#\)](#)

Remarks

This method requires that the SOLDWORKS MBD add-in be loaded. ([ISldWorks::LoadAddIn](#))

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::Capture3DView Method \(\)](#)
[IModelDocExtension::Get3DView Method \(\)](#)
[IModelDocExtension::Get3DViews Method \(\)](#)
[IView3D Interface](#)
[IModelDocExtension::Get3DViewCount Method \(\)](#)
[IModelDocExtension::Get3DViewNames Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReJogRunningDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReJogRunningDimension Method (IModelDocExtension)

Applies jogs where extension lines overlap dimension text in an angular running dimension.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReJogRunningDimension()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
instance.ReJogRunningDimension()
```

C#

```
void ReJogRunningDimension()
```

C++/CLI

```
void ReJogRunningDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReJogRunningDimension](#).

Example

[Insert Angular Running Dimension \(VBA\)](#)
[Insert Angular Running Dimension \(VB.NET\)](#)
[Insert Angular Running Dimension \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::AlignRunningDimension Method \(\)](#)
[IModelDocExtension::AddAngularRunningDim Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseStream Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReleaseStream Method (IModelDocExtension)

Releases a previously obtained stream.

.NET Syntax

Visual Basic (Declaration)

```
Function ReleaseStream( _  
    ByVal StreamType As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim StreamType As System.Integer  
Dim value As System.Boolean
```

```
value = instance.ReleaseStream(StreamType)
```

C#

```
System.bool ReleaseStream(  
    System.int StreamType  
)
```

C++/CLI

```
System.bool ReleaseStream(  
& System.int StreamType
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StreamType
1 = material stream

Return Value

True if stream is released, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReleaseStream](#).

Remarks

To obtain the stream, call [IModelDocExtension::GetStream](#).

VB Example

This example illustrates attaching an XML document to a stream, and then releasing the stream.

```
Dim swApp As Object
Sub main()
    Set swApp = Application.SldWorks
    Dim docext As SldWorks.ModelDocExtension

    Set docext = swApp.ActiveDoc.Extension

    Dim xmldoc As MSXML2.DOMDocument
    Set xmldoc = CreateObject("MSXML2.DOMDocument")
    Dim stat As Boolean
    Dim stream
    Set stream = docext.GetStream(1, stat)
    xmldoc.Load (stream)
    docext.ReleaseStream (1)

    xmldoc.Save ("C:\temp\xmlmat.xml")

End Sub
```

C++ Example

```
//-----
CComPtr<IModelDocExtension> ext;
m_iModelDoc2->get_Extension(&ext);
LPSTREAM stream = NULL;
VARIANT_BOOL access = 0;
ext->IGetStream(1, &access, &stream);
// Your code
if (stream)
{
    VARIANT_BOOL status;
```

```

    ext->IReloadStream(1, &status);
    stream->Release();
}
//-----

```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReloadOrReplace Method (IModelDocExtension)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReloadOrReplace Method (IModelDocExtension)

Reloads or replaces this model document.

.NET Syntax

Visual Basic (Declaration)

```

Function ReloadOrReplace( _
    ByVal ReadOnly As System.Boolean, _
    ByVal ReplaceFileName As System.String, _
    ByVal DiscardChanges As System.Boolean, _
    ByVal ForceReload As System.Boolean _
) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim ReadOnly As System.Boolean
Dim ReplaceFileName As System.String
Dim DiscardChanges As System.Boolean
Dim ForceReload As System.Boolean
Dim value As System.Integer

value = instance.ReloadOrReplace(ReadOnly, ReplaceFileName, DiscardChanges, ForceReload)

```

C#

```

System.int ReloadOrReplace(
    System.bool ReadOnly,
    System.string ReplaceFileName,
    System.bool DiscardChanges,
    System.bool ForceReload
)

```

C++/CLI

```

System.int ReloadOrReplace(
    & System.bool ReadOnly,
    & System.String^ ReplaceFileName,
    & System.bool DiscardChanges,
    & System.bool ForceReload
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReadOnly

True to set the model document read-only after reload or replace, false to allow write access

ReplaceFileName

Replacement model document; null or Nothing to reload the current model document

DiscardChanges

True to discard changes made to the current model document, false to abort operation if the model document was changed (see **Remarks**)

ForceReload

True to force the reload, false to not

Return Value

Error codes as defined by [swComponentReloadError_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReloadOrReplace](#).

Example

[Reload Model \(VBA\)](#)

Remarks

The model to reload or replace must be open in its own window frame.

This method does not reload or replace the model if the top assembly has been opened invisibly. In that case, make the model visible by calling [IModelDoc2::Visible](#) before calling this method.

You must call this method after calling [IModelDoc2::ForceReleaseLocks](#), which detaches a file from the file system, to re-attach a detached file to the file system. If you do not call this method after calling [IModelDoc2::ForceReleaseLocks](#), then you will experience problems with OLE objects (e.g., design tables). Any changes made to a file while it is detached are not preserved unless you save the file to disk before calling this method. Additionally if you set this method's *DiscardChanges* parameter to false and you made changes to a detached file that you are attempting to re-attach to the file system, then this method will fail. *DiscardChanges* must be set to true to re-attach a detached file.

See [ISldWorks::CloseAndReopen](#) to perform a similar function with drawing documents.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::ReloadWithReferences Method\(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReloadWithReferences Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReloadWithReferences Method (IModelDocExtension)

Reloads this model document with the specified references.

.NET Syntax

Visual Basic (Declaration)

```
Function ReloadWithReferences( _  
    ByVal ReadOnly As System.Boolean, _  
    ByVal DiscardChanges As System.Boolean, _  
    ByVal ReferencesToReload As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ReadOnly As System.Boolean  
Dim DiscardChanges As System.Boolean  
Dim ReferencesToReload As System.Object  
Dim value As System.Integer
```

```
value = instance.ReloadWithReferences(ReadOnly, DiscardChanges, ReferencesToReload)
```

C#

```
System.int ReloadWithReferences(  
    System.bool ReadOnly,  
    System.bool DiscardChanges,  
    System.object ReferencesToReload  
)
```

C++/CLI

```
System.int ReloadWithReferences(  
& System.bool ReadOnly,  
& System.bool DiscardChanges,  
& System.Object^ ReferencesToReload  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ReadOnly

True to set the model document read-only after reload or replace, false to allow write access

DiscardChanges

True to discard changes made to the current model document, false to abort operation if the model document was changed (see **Remarks**)

ReferencesToReload

Array of IDispatch pointers to referenced documents to reload

Return Value

Error codes as defined by [swComponentReloadError_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReloadWithReferences](#).

Remarks

The model to reload or replace must be open in its own window frame.

This method:

- provides a performance improvement over [IModelDocExtension::ReloadOrReplace](#) (when its ForceReload parameter is set to True).
- is faster because it reloads only a selection of referenced components.
- does not reload or replace the model if the top assembly has been opened invisibly. In that case, make the model visible by calling [IModelDoc2::Visible](#) before calling this method.
- must be called after calling [IModelDoc2::ForceReleaseLocks](#), which detaches a file from the file system, to re-attach a detached file to the file system. If you do not call this method after calling [IModelDoc2::ForceReleaseLocks](#), then you will experience problems with OLE objects (e.g., design tables). Any changes made to a file while it is detached are not preserved unless you save the file to disk before calling this method. Additionally if you set this method's DiscardChanges parameter to false and you made changes to a detached file that you are attempting to re-attach to the file system, then this method will fail. DiscardChanges must be set to true to re-attach a detached file.

See [ISldWorks::CloseAndReopen](#) to perform a similar function with drawing documents.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveMaterialProperty Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveMaterialProperty Method (IModelDocExtension)

Removes material properties from this model.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveMaterialProperty( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim Config_opt As System.Integer
Dim Config_names As System.Object
Dim value As System.Boolean

value = instance.RemoveMaterialProperty(Config_opt, Config_names)
```

C#

```
System.bool RemoveMaterialProperty(
    System.int Config\_opt,
    System.object Config\_names
)
```

C++/CLI

```
System.bool RemoveMaterialProperty(
    & System.int Config\_opt,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration options as defined by [swInConfigurationOpts_e](#)
Config_names
Array of configurations

Return Value

True if material properties successfully removed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RemoveMaterialProperty](#).

Remarks

This method:

- is intended to be used on features that have a changed material property value, for example, color.
- does not remove material. To remove material, call [IPartDoc::SetMaterialPropertyName2](#), passing an empty string to the Name parameter.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::IGetMaterialPropertyValues Method](#)
[IModelDocExtension::IRemoveMaterialProperty Method](#)
[IModelDocExtension::ISetMaterialPropertyValues Method](#)
[IModelDocExtension::SetMaterialPropertyValues Method](#)
[IModelDocExtension::HasMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveModelColorizer Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveModelColorizer Method (IModelDocExtension)

Removes your installed implemented interface of the [ISwColorContour](#) interface.

 .NET Syntax

Visual Basic (Declaration)

Sub RemoveModelColorizer(_
 ByVal [PInterface](#) As System.Object _
)

Visual Basic (Usage)

Dim instance As [IModelDocExtension](#)
Dim PInterface As System.Object

instance.RemoveModelColorizer(PInterface)

C#

void RemoveModelColorizer(
 System.object [PInterface](#)
)

C++/CLI

void RemoveModelColorizer(
 & System.Object^ [PInterface](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PInterface
Pointer to your installed implemented interface

 Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RemoveModelColorizer](#).

 See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::InstallModelColorizer Method](#)

 Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveTexture Method (IModelDocExtension)

[See Also](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveTexture Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::RemoveTexture2](#).

 .NET Syntax

Visual Basic (Declaration)

Function RemoveTexture() As System.Boolean

Visual Basic (Usage)

Dim instance As [IModelDocExtension](#)
Dim value As System.Boolean

```
value = instance.RemoveTexture()
```

C#

```
System.bool RemoveTexture()
```

C++/CLI

```
System.bool RemoveTexture();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RemoveTexture](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

RemoveTexture2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveTexture2 Method (IModelDocExtension)

Removes the texture from the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTexture2( _  
    ByVal Config\_name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Config_name As System.String  
Dim value As System.Boolean
```

```
value = instance.RemoveTexture2(Config_name)
```

C#

```
System.bool RemoveTexture2(  
    System.string Config\_name  
)
```

C++/CLI

```
System.bool RemoveTexture2(  
& System.String^ Config\_name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_name
Name of the configuration

Return Value

True if the texture is removed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RemoveTexture2](#).

Example

[Remove Textures from Assembly Components \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CreateTexture Method](#)
[IModelDocExtension::SetTexture Method](#)
[IModelDocExtension::GetTexture Method](#)
[IModelDocExtension::SetTextureByDisplayState Method](#)
[IModelDocExtension::RemoveTextureByDisplayState Method](#)
[IFace2::RemoveTexture2 Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveTextureByDisplayState Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveTextureByDisplayState Method (IModelDocExtension)

Removes the texture applied to this model in the specified display state.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTextureByDisplayState( _  
    ByVal DisplayStateName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim DisplayStateName As System.String  
Dim value As System.Boolean  
  
value = instance.RemoveTextureByDisplayState(DisplayStateName)
```

C#

```
System.bool RemoveTextureByDisplayState(  
    System.string DisplayStateName  
)
```

C++/CLI

```
System.bool RemoveTextureByDisplayState(  
& System.String^ DisplayStateName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayStateName
Name of display state

Return Value

True if texture removed from model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.RemoveTextureByDisplayState](#).

Example

[Apply and Remove Texture To and From Model By Display State \(C#\)](#)
[Apply and Remove Texture To and From Model By Display State \(VB.NET\)](#)
[Apply and Remove Texture To and From Model By Display State \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetTexture Method](#)
[IModelDocExtension::RemoveTexture2 Method](#)
[IModelDocExtension::SetTexture Method](#)
[IModelDocExtension::SetTextureByDisplayState Method](#)
[IModelDocExtension::CreateTexture Method](#)

Availability

SOLIDWORKS 2010 SP1, Revision Number 18.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveVisibleBox Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RemoveVisibleBox Method (IModelDocExtension)

Removes the visible bounding box set by [IModelDocExtension::SetVisibleBox](#) and resets the size of the bounding box to the size calculated by SOLIDWORKS for a part or an assembly.

.NET Syntax

Visual Basic (Declaration)

```
Sub RemoveVisibleBox()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
instance.RemoveVisibleBox()
```

C#

```
void RemoveVisibleBox()
```

C++/CLI

```
void RemoveVisibleBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RemoveVisibleBox](#).

Example

[Set Visible Bounding Box for Zoom to Fit \(C#\)](#)
[Set Visible Bounding Box for Zoom to Fit \(VB.NET\)](#)
[Set Visible Bounding Box for Zoom to Fit \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetVisibleBox Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RenameDocument Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RenameDocument Method (IModelDocExtension)

Temporarily renames the selected component using the specified name.

.NET Syntax

Visual Basic (Declaration)

```
Function RenameDocument( _  
    ByVal NewName As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim NewName As System.String  
Dim value As System.Integer  
  
value = instance.RenameDocument(NewName)
```

C#

```
System.int RenameDocument(  
    System.string NewName  
)
```

C++/CLI

```
System.int RenameDocument(  
& System.String^ NewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewName
New name for the component

Return Value

Status of temporarily renaming the component as defined in [swRenameDocumentError_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RenameDocument](#).

Example

[Rename Components and Save Assembly \(C#\)](#)
[Rename Components and Save Assembly \(VB.NET\)](#)
[Rename Components and Save Assembly \(VBA\)](#)
[Rename Component and Update References \(C#\)](#)
[Rename Component and Update References \(VB.NET\)](#)
[Rename Component and Update References \(VBA\)](#)

Remarks

If this method returns `swRenameDocumentError_e.swRenameDocumentError_None`, then the new name of the component is shown in the FeatureManager design tree and the file name of the component changes in memory. All currently open documents that reference the renamed file are updated to reference the new file name.

To:

- get whether the document has renamed components, call [IModelDocExtension::HasRenamedDocuments](#).
- to avoid an error when attempting to save the document without first saving its references, use [IRenamedDocumentReferences](#) to update references to the renamed component in unopened documents.
- permanently rename the component, [save](#) the document.

See the SOLIDWORKS Help for details about renaming components.

See Also

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RenameDraftingStandard Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RenameDraftingStandard Method (IModelDocExtension)

Rename the current custom drafting to the specified name.

.NET Syntax

Visual Basic (Declaration)

```
Function RenameDraftingStandard( _  
    ByVal Name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim value As System.Boolean  
  
value = instance.RenameDraftingStandard(Name)
```

C#

```
System.bool RenameDraftingStandard(  
    System.string Name  
)
```

C++/CLI

```
System.bool RenameDraftingStandard(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

New name for current custom drafting standard

Return Value

True if the current custom drafting standard is renamed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RenameDraftingStandard](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CopyDraftingStandard Method](#)
[IModelDocExtension::DeleteDraftingStandard Method](#)
[IModelDocExtension::GetDraftingStandardNames Method](#)
[IModelDocExtension::LoadDraftingStandard Method](#)
[IModelDocExtension::SaveDraftingStandard Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReorderFeature Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReorderFeature Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::ReorderFeature2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ReorderFeature( _  
    ByVal FeatureToMove As System.String, _  
    ByVal TargetFeature As System.String, _  
    ByVal MoveLocation As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FeatureToMove As System.String  
Dim TargetFeature As System.String  
Dim MoveLocation As System.Integer  
Dim value As System.Boolean  
  
value = instance.ReorderFeature(FeatureToMove, TargetFeature, MoveLocation)
```

C#

```
System.bool ReorderFeature(  
    System.string FeatureToMove,  
    System.string TargetFeature,  
    System.int MoveLocation  
)
```

C++/CLI

```
System.bool ReorderFeature(  
& System.String^ FeatureToMove,  
& System.String^ TargetFeature,  
& System.int MoveLocation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureToMove

Name of feature to move

TargetFeature

Name of feature before or after which to move FeatureToMove; valid only if MoveLocation is [swMoveLocation_e.swMoveAfter](#)

- OR -

Name of folder; valid only if MoveLocation is swMoveLocation_e.swMoveToFolder

MoveLocation

Move type as defined by [swMoveLocation_e](#)

Return Value

True if feature moved successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReorderFeature](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IPartDoc::ReorderFeature Method \(\)](#)

[IAssemblyDoc::ReorderComponents Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReorderFeature2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReorderFeature2 Method (IModelDocExtension)

Moves the specified feature to another location in the FeatureManager design tree of this part or assembly.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function ReorderFeature2( _  
    ByVal FeatToMove As System.String, _  
    ByVal TargetFeat As System.String, _  
    ByVal MoveLocation As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FeatToMove As System.String  
Dim TargetFeat As System.String  
Dim MoveLocation As System.Integer  
Dim value As System.Boolean  
  
value = instance.ReorderFeature2(FeatToMove, TargetFeat, MoveLocation)
```

C#

```
System.bool ReorderFeature2(  
    System.string FeatToMove,  
    System.string TargetFeat,  
    System.int MoveLocation  
)
```

C++/CLI

```
System.bool ReorderFeature2(  
& System.String^ FeatToMove,  
& System.String^ TargetFeat,  
& System.int MoveLocation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatToMove

Name of feature to move (see **Remarks**)

TargetFeat

Name of feature before or after which to move FeatureToMove

- Or -

Name of folder; valid only if MoveLocation is swMoveLocation_e.swMoveToFolder and FeatToMove is not a Cut list/Solid Bodies folder (see **Remarks**)

MoveLocation

Move type as defined by [swMoveLocation_e](#) (see **Remarks**)

Return Value

True if feature moved successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReorderFeature2](#).

Example

'VBA

'This example demonstrates how to use this method to re-order folders in the Cut list/Solid Bodies folder.

' Preconditions:

' 1. Open *Public_documents\samples\tutorial\api\weldment_box3.sldprt*.

' 2. Open an Immediate window.

' 3. Run/Debug the macro.

'

'Postconditions:

' 1. When the macro pauses, inspect the Cut list folder.

' 2. Observe the order of two new sub folders in the Cut list folder: **Sub-weldment2** comes before **Sub-weldment3**.

' 3. Press F5 to reorder these folders.

' 4. When the macro pauses, inspect the reordered folders: **Sub-weldment3** moves before **Sub-weldment2**.

' 5. Select a body or folder in the Cut list folder.

' 6. Press F5 to locate the selection's parent folder.

' 7. Inspect the Immediate window for the parent folder type and name (Type = "SolidBodyFolder" and Name = "Solid Bodies").

'

' NOTE: Do not save the model as it is used by other examples.

'=====

```
Dim swApp As SldWorks.SldWorks
```

```
Dim myFeature As SldWorks.Feature
```

```
Dim Part As SldWorks.ModelDoc2
```

```
Dim swFolderFeat As SldWorks.Feature
```

```
Dim swFeatureMgr As SldWorks.FeatureManager
```

```
Dim swSelectionMgr As SldWorks.SelectionMgr
```

```
Dim swSelObj As Object
```

```
Dim boolstatus As Boolean
```

```
Dim longstatus As Long, longwarnings As Long
```

```
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Set Part = swApp.ActiveDoc
```

```
    boolstatus = Part.Extension.SelectByID2("Structural Member1[1]", "SOLIDBODY", 0,  
0, 0, False, 0, Nothing, 0)
```

```
    boolstatus = Part.Extension.SelectByID2("Structural Member1[2]", "SOLIDBODY", 0,  
0, 0, True, 0, Nothing, 0)
```

```
    boolstatus = Part.Extension.SelectByID2("Structural Member1[3]", "SOLIDBODY", 0,  
0, 0, True, 0, Nothing, 0)
```

```
    boolstatus = Part.Extension.SelectByID2("Structural Member1[4]", "SOLIDBODY", 0,  
0, 0, True, 0, Nothing, 0)
```

```

    boolstatus = Part.Extension.SelectByID2("Structural Member1[6]", "SOLIDBODY", 0,
0, 0, True, 0, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Structural Member1[8]", "SOLIDBODY", 0,
0, 0, True, 0, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Structural Member3[1]", "SOLIDBODY", 0,
0, 0, True, 0, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Structural Member3[2]", "SOLIDBODY", 0,
0, 0, True, 0, Nothing, 0)

    Set myFeature = Part.FeatureManager.InsertSubFolder()

    boolstatus = Part.Extension.SelectByID2("Gusset1", "SOLIDBODY", 0, 0, 0, False,
0, Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Gusset2", "SOLIDBODY", 0, 0, 0, True, 0,
Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Gusset3", "SOLIDBODY", 0, 0, 0, True, 0,
Nothing, 0)
    boolstatus = Part.Extension.SelectByID2("Gusset4", "SOLIDBODY", 0, 0, 0, True, 0,
Nothing, 0)

    Set myFeature = Part.FeatureManager.InsertSubFolder()

    Stop '1. Inspect the order of cut list folders;
    '2. Press F5 to reorder the folders

    boolstatus = Part.Extension.ReorderFeature2("Sub-weldment2", "Sub-weldment3",
swMoveLocation_e.swMoveAfter)
    Stop '1. Inspect the new order of cut list folders;
    '2. Select a body or body folder in the cut list folder;
    '3. Press F5 to locate its parent folder

    Set swSelectionMgr = Part.SelectionManager
    Debug.Print "Selected Obj Type: " & swSelectionMgr.GetSelectedObjectType3(1, -1)
    Set swSelObj = swSelectionMgr.GetSelectedObject6(1, -1)

    ' Locate the parent folder of the selection
    Set swFeatureMgr = Part.FeatureManager
    Set swFolderFeat = swFeatureMgr.CutListFolderLocation(swSelObj)

    If swFolderFeat Is Nothing Then
        Debug.Print "Please select a solid body/cut list item or solid body/cut list
folder"
        Exit Sub
    End If

```

```
Debug.Print "Parent folder Type: " & swFolderFeat.GetTypeName2 & " Name: " & swFolderFeat.Name
```

End Sub

 Example

[Reorder Features \(VBA\)](#)
[Reorder Features \(VB.NET\)](#)
[Reorder Features \(C#\)](#)

 Remarks

This method functions just like the now obsolete `IModelDocExtension::ReorderFeature`, but also supports reordering of cut list folders and cut list sub-weldment folders.

When reordering cut list folders and sub-weldment folders, the following restrictions apply:

- `FeatToMove` is a folder feature: cut list / solid body / sub-weldment folder; `FeatToMove` is not a cut list item/solid body.
- This method:
 - reorders solid folders in parts and assembly components.
 - is called after assigning a `ModelDoc` variable to the part or assembly component whose cut list folders you want to reorder.
 - only affects cut lists in the active configuration. To reorder a cut list folder in another configuration, you must make it active before calling this method.
 - returns false if you try to move of a cut list/solid body folder to the top using either `MoveLocation = swMoveLocation_e.swMoveToTop` or `MoveLocation = swMoveLocation_e.swMoveBefore`. For example, if `TargetFeat` is the top folder, then you cannot set `MoveLocation` to `swMoveLocation_e.swMoveBefore`.

Use [IBodyFolder::GetTopLevelFeatureCount](#) and [IBodyFolder::GetTopLevelFeatures](#) to obtain top level folders in a body folder.

 See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IPartDoc::ReorderFeature Method \(\)](#)
[IAssemblyDoc::ReorderComponents Method \(\)](#)

 Availability

SOLIDWORKS 2025 SP02, Revision Number 33.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

ResetStandardViews Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ResetStandardViews Method (IModelDocExtension)

Returns all standard model views to their default settings.

 .NET Syntax

Visual Basic (Declaration)

```
Function ResetStandardViews() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
value = instance.ResetStandardViews()
```

C#

```
System.bool ResetStandardViews()
```

C++/CLI

```
System.bool ResetStandardViews();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True to return all standard model views to their default settings, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ResetStandardViews](#).

Remarks

This method works with parts and assemblies only.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::UpdateStandardViews Method \(\)](#)
[IModelDoc2::ShowNamedView2 Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReverseDecalsOrder Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ReverseDecalsOrder Method (IModelDocExtension)

Reverses the order of the decals on the model.

.NET Syntax

Visual Basic (Declaration)

```
Function ReverseDecalsOrder( _  
    ByVal DecalID As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim DecalID As System.Integer  
Dim value As System.Boolean  
  
value = instance.ReverseDecalsOrder(DecalID)
```

C#

```
System.bool ReverseDecalsOrder(  
    System.int DecalID  
)
```

C++/CLI

```
System.bool ReverseDecalsOrder(  
& System.int DecalID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DecalID
[Decal ID](#)

Return Value

True if the order of the decals is reversed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ReverseDecalsOrder](#).

Example

[Reverse Order of Decals \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDecal Interface](#)
[IModelDocExtension::AddDecal Method](#)
[IModelDocExtension::CreateDecal Method](#)
[IModelDocExtension::DeleteAllDecals Method](#)
[IModelDocExtension::DeleteDecal Method](#)
[IModelDocExtension::GetDecal Method](#)
[IModelDocExtension::GetDecals Method](#)
[IModelDocExtension::GetDecalsCount Method](#)
[IModelDocExtension::HideDecal Method](#)
[IModelDocExtension::IGetDecals Method](#)
[IModelDocExtension::MoveDecal Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateOrCopy Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RotateOrCopy Method (IModelDocExtension)

Rotates and optionally copies the selected sketch entities or annotations.

.NET Syntax

Visual Basic (Declaration)

```
Sub RotateOrCopy( _  
    ByVal Copy As System.Boolean, _  
    ByVal NumCopies As System.Integer, _  
    ByVal KeepRelations As System.Boolean, _  
    ByVal BaseX As System.Double, _  
    ByVal BaseY As System.Double, _  
    ByVal BaseZ As System.Double, _  
    ByVal DestX As System.Double, _  
    ByVal DestY As System.Double, _  
    ByVal DestZ As System.Double, _  
    ByVal Angle As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Copy As System.Boolean  
Dim NumCopies As System.Integer  
Dim KeepRelations As System.Boolean  
Dim BaseX As System.Double  
Dim BaseY As System.Double  
Dim BaseZ As System.Double  
Dim DestX As System.Double  
Dim DestY As System.Double  
Dim DestZ As System.Double  
Dim Angle As System.Double
```

```
instance.RotateOrCopy(Copy, NumCopies, KeepRelations, BaseX, BaseY, BaseZ, DestX, DestY, DestZ,  
Angle)
```

C#

```
void RotateOrCopy(  
    System.bool Copy,  
    System.int NumCopies,  
    System.bool KeepRelations,
```

```

        System.double BaseX,
        System.double BaseY,
        System.double BaseZ,
        System.double DestX,
        System.double DestY,
        System.double DestZ,
        System.double Angle
    )

```

C++/CLI

```

void RotateOrCopy(
    & System.bool Copy,
    & System.int NumCopies,
    & System.bool KeepRelations,
    & System.double BaseX,
    & System.double BaseY,
    & System.double BaseZ,
    & System.double DestX,
    & System.double DestY,
    & System.double DestZ,
    & System.double Angle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Copy

True to copy sketch entities or annotations, false to not

NumCopies

Number of copies

KeepRelations

True to keep sketch relations, false to not

BaseX

X coordinate of the base point from which to rotate the sketch entities or annotations

BaseY

Y coordinate of the base point from which to rotate the sketch entities or annotations

BaseZ

Z coordinate of the base point from which to rotate the sketch entities or annotations

DestX

X vector component of the axis of rotation parallel to the base point

DestY

Y vector component of the axis of rotation parallel to the base point

DestZ

Z vector component of the axis of rotation parallel to the base point

Angle

Angle at which to rotate or move the sketch entities or annotations

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RotateOrCopy](#).

Example

[Rotate, Scale, and Copy Sketch \(C#\)](#)

[Rotate, Scale, and Copy Sketch \(VB.NET\)](#)

[Rotate, Scale, and Copy Sketch \(VBA\)](#)

Remarks

Typically a user will specify 0, 0, 1 for the x, y, z destination point coordinates and 0 for the baseZ argument. The baseX and baseY arguments should specify the center of rotation for the sketch entity.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::MoveOrCopy Method](#)

[IModelDocExtension::ScaleOrCopy Method](#)

Availability

RunCommand Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : RunCommand Method (IModelDocExtension)

Runs the specified SOLIDWORKS command.

.NET Syntax

Visual Basic (Declaration)

```
Function RunCommand( _  
    ByVal CommandID As System.Integer, _  
    ByVal NewTitle As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim CommandID As System.Integer  
Dim NewTitle As System.String  
Dim value As System.Boolean  
  
value = instance.RunCommand(CommandID, NewTitle)
```

C#

```
System.bool RunCommand(  
    System.int CommandID,  
    System.string NewTitle  
)
```

C++/CLI

```
System.bool RunCommand(  
& System.int CommandID,  
& System.String^ NewTitle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CommandID

SOLIDWORKS command as defined in [swCommands_e](#)

NewTitle

Your title for the SOLIDWORKS PropertyManager page, if invoked by this command

Return Value

True if the SOLIDWORKS command ran, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::RunCommand](#).

Example

[Run SOLIDWORKS Commands \(VBA\)](#)

[Fire Events When PropertyManager Page Opened and Canceled \(C#\)](#)

[Fire Events When PropertyManager Page Opened and Canceled \(VB.NET\)](#)

[Fire Events When PropertyManager Page Opened and Canceled \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[ISldWorks::RunCommand Method](#)

[ISldWorks::IsCommandEnabled Method](#)

Availability

SaveAs Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveAs Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::SaveAs2](#)

.NET Syntax

Visual Basic (Declaration)

```

Function SaveAs( _
    ByVal Name As System.String, _
    ByVal Version As System.Integer, _
    ByVal Options As System.Integer, _
    ByVal ExportData As System.Object, _
    ByRef Errors As System.Integer, _
    ByRef Warnings As System.Integer _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim Name As System.String
Dim Version As System.Integer
Dim Options As System.Integer
Dim ExportData As System.Object
Dim Errors As System.Integer
Dim Warnings As System.Integer
Dim value As System.Boolean

value = instance.SaveAs(Name, Version, Options, ExportData, Errors, Warnings)

```

C#

```

System.bool SaveAs(
    System.string Name,
    System.int Version,
    System.int Options,
    System.object ExportData,
    out System.int Errors,
    out System.int Warnings
)

```

C++/CLI

```

System.bool SaveAs(
    & System.String^ Name,
    & System.int Version,
    & System.int Options,
    & System.Object^ ExportData,
    [Out] System.int Errors,
    [Out] System.int Warnings
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Full pathname of the document to save; the file extension indicates any conversion that should be performed (for example, **Part1.igs** to save in IGES format) (see **Remarks**)

Version

Format in which to save this document as defined in [swSaveAsVersion_e](#) (see **Remarks**)

Options

Option indicating how to save the document as defined in [swSaveAsOptions_e](#) (see **Remarks**)

ExportData

[IExportPdfData](#) object for exporting drawing sheets to PDF (see **Remarks**)

Errors

Errors that caused the save to fail as defined in [swFileSaveError_e](#) (see **Remarks**)

Warnings

Warnings or extra information generated during the save operation as defined in [swFileSaveWarning_e](#) (see **Remarks**)

Return Value

True if the save is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveAs](#).

Example

[Save File As PDF \(VBA\)](#)
[Save File as PDF \(C#\)](#)
[Save File as PDF \(VB.NET\)](#)
[Import STEP File \(C#\)](#)
[Import STEP File \(VB.NET\)](#)
[Import STEP File \(VBA\)](#)

Remarks

To save as an IGES, STL, or STEP file, the document to convert must be the active document. Before calling this method:

1. Call [ISldWorks::ActivateDoc3](#) to make the document to convert the active document.
2. Call [ISldWorks::ActiveDoc](#) to get the active document.

This method:

- Exports the entire model, unless faces or bodies are selected, in which case, it exports only those. Call [IModelDoc2::ClearSelection2](#) before calling IModelDocExtension::SaveAs to clear the selection list and export the entire model.
- Overwrites existing files unless they are read only.
- Results in the FileSaveNotify event being sent to any application listening.
- Removes any configuration-specific bitmap previews, except the current configuration's.

Saving a document as PDF when the document is open as view only is not supported.

Do not use ModelDocExtension::SaveAs to copy assemblies, drawings, or parts with in-context references. Instead, use [ISldWorks::CopyDocument](#) or [ISldWorks::ICopyDocument](#).

Use Name to specify the full pathname of the saved document. If you specify only the file name, then it is saved in the active document's directory. The filename extension indicates the conversion that should be performed (for example, **Part1.igs** to save to IGES). If the filename extension does not uniquely indicate how the file should be formatted, use Version to specify how to save the file. For example, to save:

- A standard drawing document as a detached drawing, specify swSaveAsDetachedDrawing for Version.
- A detached drawing as a standard drawing, specify swSaveAsStandardDrawing for Version.
- A standard or detached drawing document in the same format, specify swSaveAsCurrentVersion for Version.

Use ExportData to specify which drawings sheets to save to PDF. If ExportData is Nothing or null, then all sheets are saved to PDF. Saving a document as PDF when the document is open as view-only is not supported.

Use Options to specify save options. You can specify additional options using [ISldWorks::SetUserPreferenceIntegerValue](#). For example:

```
' Save assembly as multibody part and save exterior faces as surface bodies
```

```
swApp.SetUserPreferenceIntegerValue swSaveAssemblyAsPartOptions, _  
    swSaveAsmAsPart_ExteriorFaces  
swModelDocExt.SaveAs "H:\Assem1.SLDPRPRT", swSaveAsCurrentVersion, _  
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

- Or -

```
' Save all drawing sheets in active drawing document as an eDrawings file  
swApp.SetUserPreferenceIntegerValue swEdrawingsSaveAsSelectionOption, swEdrawingSaveAll  
swModelDocExt.SaveAs "H:\Grid.edrw", swSaveAsCurrentVersion, _  
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

If the file is saved successfully, then the returned value is true and Errors is 0. If the save is not successful, then the returned value is false and Errors contains a bitwise OR of the error codes that were generated in saving the document. Check the masks against the [swFileSaveError_e](#) enumeration. If you do not want SOLIDWORKS to return error information, set Errors to Nothing or null.

Even if the file is saved successfully, there might be warnings or information that occur during the save operation in which you might be interested. Warnings contains a bitwise OR of the warning codes that were generated when saving the document. Check the masks against the [swFileSaveWarning_e](#) enumeration. If you do not want warning information returned, set Warnings to Nothing or null.

Use [IModelDoc2::Save3](#) to save a file using its current name.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[ISldWorks::ActivateDoc2 Method](#)
[ISldWorks::CheckpointConvertedDocument Method](#)
[ISldWorks::CloseAllDocuments Method](#)
[ISldWorks::CloseDoc Method](#)
[ISldWorks::CopyDocument Method](#)
[ISldWorks::ExitApp Method](#)
[ISldWorks::LoadFile4 Method](#)
[ISldWorks::OpenDoc6 Method](#)
[ISldWorks::QuitDoc Method](#)
[ISldWorks::iActiveDoc2 Property](#)
[ISldWorks::ActiveDoc Property](#)
[IModelDocExtension::SaveDeFeaturedFile Method](#)

Availability

SOLIDWORKS 2007 SP1, Revision Number 15.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

SaveAs2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveAs2 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::SaveAs3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SaveAs2( _  
    ByVal Name As System.String, _  
    ByVal Version As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal ExportData As System.Object, _  
    ByVal ReferencePrefixOrSuffixText As System.String, _  
    ByVal AddTextAsPrefix As System.Boolean, _  
    ByRef Errors As System.Integer, _  
    ByRef Warnings As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim Version As System.Integer  
Dim Options As System.Integer  
Dim ExportData As System.Object  
Dim ReferencePrefixOrSuffixText As System.String  
Dim AddTextAsPrefix As System.Boolean  
Dim Errors As System.Integer  
Dim Warnings As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SaveAs2(Name, Version, Options, ExportData, ReferencePrefixOrSuffixText,  
AddTextAsPrefix, Errors, Warnings)
```

C#

```
System.bool SaveAs2(  
    System.string Name,  
    System.int Version,  
    System.int Options,  
    System.object ExportData,  
    System.string ReferencePrefixOrSuffixText,  
    System.bool AddTextAsPrefix,  
    out System.int Errors,  
    out System.int Warnings)
```

```
        out System.int Warnings
    )
```

C++/CLI

```
System.bool SaveAs2(
& System.String^ Name,
& System.int Version,
& System.int Options,
& System.Object^ ExportData,
& System.String^ ReferencePrefixOrSuffixText,
& System.bool AddTextAsPrefix,
& [Out] System.int Errors,
& [Out] System.int Warnings
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Full pathname of the document to save; the file extension indicates any conversion that should be performed (for example, **Part1.igs** to save in IGES format) (see **Remarks**)

Version

Format in which to save this document as defined in [swSaveAsVersion_e](#) (see **Remarks**)

Options

Option indicating how to save the document as defined in [swSaveAsOptions_e](#) (see **Remarks**)

ExportData

[IExportPdfData](#) object for exporting drawing sheets to PDF (see **Remarks**)

ReferencePrefixOrSuffixText

Text to prefix names of reference files (AddTextAsPrefix is true) or text to suffix them (AddTextAsPrefix is false); empty string to neither prefix nor suffix the names of reference files; valid only for assemblies (see **Remarks**)

AddTextAsPrefix

True to prefix names of reference files with ReferencePrefixOrSuffixText, false to suffix names of reference files with ReferencePrefixOrSuffixText; valid only for assemblies and when ReferencePrefixOrSuffixText is a non-empty string (see **Remarks**)

Errors

Errors that caused the save to fail as defined in [swFileSaveError_e](#) (see **Remarks**)

Warnings

Warnings or extra information generated during the save operation as defined in [swFileSaveWarning_e](#) (see **Remarks**)

Return Value

True if the save is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveAs2](#).

Example

'VBA

'Preconditions:

'1. Open **C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\tutorial\advdrawings\98food processor.sldasm**.

'2. Ensure **c:\temp** exists.

'Postconditions:

'1. Silently saves a copy of the assembly (**food_processor_Suff.sldasm**) and re-named reference files (***_Suff.sldprt**) to **c:\temp**.

'2. Opens the saved assembly.

'3. Observe the re-named components in the FeatureManager design tree.

' NOTE: Because the model is used elsewhere, do not save changes.

'=====

Option Explicit

```
Dim swApp           As SldWorks.SldWorks
Dim swModel          As SldWorks.ModelDoc2
Dim swModelDocExt    As SldWorks.ModelDocExtension
Dim boolstatus       As Boolean
Dim FileName         As String
Dim DIR              As String
Dim EXT              As String
Dim opt              As Long
Dim IErrors           As Long
Dim IWarnings        As Long
```

Sub main()

```

DIR = "c:\temp\"

EXT = ".SLDASM"
Set swApp = Application.SldWorks
swApp.Visible = True

Set swModel = swApp.ActiveDoc
Set swModelDocExt = swModel.Extension

'1 = swSaveAsOptions_Silent
'4 = swSaveAsOptions_SaveReferenced
'512 = swSaveAsOptions_CopyAndOpen

opt = 512 + 4 + 1

FileName = DIR & "food_processor_Suff" & EXT

boolstatus = swModelDocExt.SaveAs2(FileName, 0, opt, Nothing, "_Suff", False, IErrors, IWarnings)

End Sub

```

Remarks

The difference between this method and the now obsolete [IModelDocExtension::SaveAs](#) is that when you save assemblies using this method, you can prefix or suffix reference file names by specifying `AddTextAsPrefix` and `ReferencePrefixOrSuffixText`. To save references in the assembly specify Options by bitwise ANDing `swSaveAsOptions_e`:

- `swSaveAsOptions_SaveReferenced`
- `swSaveAsOptions_Copy` or `swSaveAsOptions_CopyAndOpen`
- `swSaveAsOptions_Silent`

To save as an IGES, STL, or STEP file, the document to convert must be the active document. Before calling this method:

1. Call [ISldWorks::ActivateDoc3](#) to make the document to convert the active document.
2. Call [ISldWorks::ActiveDoc](#) to get the active document.

This method:

- Exports the entire model, unless faces or bodies are selected, in which case, it exports only those. Call [IModelDoc2::ClearSelection2](#) before calling `IModelDocExtension::SaveAs` to clear the selection list and export the entire model.
- Overwrites existing files unless they are read only.
- Results in the `FileSaveNotify` event being sent to any application listening.
- Removes any configuration-specific bitmap previews, except the current configuration's.

Saving a document as PDF when the document is open as view only is not supported.

Do not use `ModelDocExtension::SaveAs` to copy assemblies, drawings, or parts with in-context references. Instead, use [ISldWorks::CopyDocument](#) or [ISldWorks::ICopyDocument](#).

Use `Name` to specify the full pathname of the saved document. If you specify only the file name, then it is saved in the active document's directory. The filename extension indicates the conversion that should be performed (for example, **Part1.igs** to save to IGES). If the filename extension does not uniquely indicate how the file should be formatted, use `Version` to specify how to save the file. For example, to save:

- A standard drawing document as a detached drawing, specify `swSaveAsDetachedDrawing` for `Version`.
- A detached drawing as a standard drawing, specify `swSaveAsStandardDrawing` for `Version`.
- A standard or detached drawing document in the same format, specify `swSaveAsCurrentVersion` for `Version`.

Use `ExportData` to specify which drawings sheets to save to PDF. If `ExportData` is `Nothing` or `null`, then all sheets are saved to PDF. Saving a document as PDF when the document is open as view-only is not supported.

Use `Options` to specify save options. You can specify additional options using [ISldWorks::SetUserPreferenceIntegerValue](#). For example:

```
' Save assembly as multibody part and save exterior faces as surface bodies
```

```
swApp.SetUserPreferenceIntegerValue swSaveAssemblyAsPartOptions, _
    swSaveAsmAsPart_ExteriorFaces
```

```
swModelDocExt.SaveAs "H:\Assem1.SLDPRT", swSaveAsCurrentVersion, _
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

- or -

```
' Save all drawing sheets in active drawing document as an eDrawings file
swApp.SetUserPreferenceIntegerValue swEdrawingsSaveAsSelectionOption, swEdrawingSaveAll
swModelDocExt.SaveAs "H:\Grid.edrw", swSaveAsCurrentVersion, _
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

If the file is saved successfully, then the returned value is true and Errors is 0. If the save is not successful, then the returned value is false and Errors contains a bitwise OR of the error codes that were generated in saving the document. Check the masks against the [swFileSaveError_e](#) enumeration. If you do not want SOLIDWORKS to return error information, set Errors to Nothing or null.

Even if the file is saved successfully, there might be warnings or information that occur during the save operation in which you might be interested. Warnings contains a bitwise OR of the warning codes that were generated when saving the document. Check the masks against the [swFileSaveWarning_e](#) enumeration. If you do not want warning information returned, set Warnings to Nothing or null.

Use [IModelDoc2::Save3](#) to save a file using its current name.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SaveDeFeaturedFile Method \(\)](#)
[ISldWorks::ActivateDoc3 Method \(\)](#)
[ISldWorks::ActiveDoc Property \(\)](#)
[ISldWorks::CheckpointConvertedDocument Method \(\)](#)
[ISldWorks::CloseAllDocuments Method \(\)](#)
[ISldWorks::CloseDoc Method \(\)](#)
[ISldWorks::CopyDocument Method \(\)](#)
[ISldWorks::ExitApp Method \(\)](#)
[ISldWorks::LoadFile4 Method \(\)](#)
[ISldWorks::OpenDoc7 Method \(\)](#)
[ISldWorks::QuitDoc Method \(\)](#)

Availability

SOLIDWORKS 2019 SP01, Revision Number 27.1

SOLIDWORKS API Help

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SaveAs3 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveAs3 Method (IModelDocExtension)

Saves the active document to the specified name with advanced options.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveAs3( _  
    ByVal Name As System.String, _  
    ByVal Version As System.Integer, _  
    ByVal Options As System.Integer, _  
    ByVal ExportData As System.Object, _  
    ByVal AdvancedSaveAsOptions As System.Object, _  
    ByRef Errors As System.Integer, _  
    ByRef Warnings As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim Version As System.Integer  
Dim Options As System.Integer  
Dim ExportData As System.Object  
Dim AdvancedSaveAsOptions As System.Object  
Dim Errors As System.Integer  
Dim Warnings As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SaveAs3(Name, Version, Options, ExportData, AdvancedSaveAsOptions, Errors,  
Warnings)
```

C#

```
System.bool SaveAs3(  
    System.string Name,  
    System.int Version,  
    System.int Options,  
    System.object ExportData,  
    System.object AdvancedSaveAsOptions,  
    out System.int Errors,  
    out System.int Warnings  
)
```

C++/CLI

```
System.bool SaveAs3(  
& System.String^ Name,  
& System.int Version,  
& System.int Options,  
& System.Object^ ExportData,  
& System.Object^ AdvancedSaveAsOptions,  
& [Out] System.int Errors,  
& [Out] System.int Warnings  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Full pathname of the document to save; the file extension indicates any conversion that should be performed (for example, **Part1.igs** to save in IGES format) (see **Remarks**)

Version

Format in which to save this document as defined in [swSaveAsVersion_e](#) (see **Remarks**)

Options

Option indicating how to save the document as defined in [swSaveAsOptions_e](#) (see **Remarks**)

ExportData

[IExportPdfData](#) object for exporting drawing sheets to PDF (see **Remarks**)

AdvancedSaveAsOptions

[IAdvancedSaveAsOptions](#) (see **Remarks**)

Errors

Errors that caused the save to fail as defined in [swFileSaveError_e](#) (see **Remarks**)

Warnings

Warnings or extra information generated during the save operation as defined in [swFileSaveWarning_e](#) (see **Remarks**)

Return Value

True if the save is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveAs3](#).

Example

See the [IAdvancedSaveAsOptions](#) example.

Remarks

The difference between this method and the now obsolete [IModelDocExtension::SaveAs2](#) is that this method uses an [IAdvancedSaveAsOptions](#) object to specify advanced options. For example:

- Saving a subset of configurations
- Renaming and/or relocating individual component references
- Specifying the prefix and/or suffix for all component reference names

To specify the [AdvancedSaveAsOptions](#) parameter of this method, call [IModelDocExtension::GetAdvancedSaveAsOptions](#) to create an [IAdvancedSaveAsOptions](#) object.

To save as an IGES, STL, or STEP file, the document to convert must be the active document. Before calling this method:

1. Call [ISldWorks::ActivateDoc3](#) to make the document to convert the active document.
2. Call [ISldWorks::ActiveDoc](#) to get the active document.

This method:

- Exports the entire model, unless faces or bodies are selected, in which case, it exports only those. Call [IModelDoc2::ClearSelection2](#) before calling this method to clear the selection list and export the entire model.
- Overwrites existing files unless they are read only.
- Results in the `FileSaveNotify` event being sent to any application listening.
- Removes any configuration-specific bitmap previews, except the current configuration's.

Use Name to specify the full pathname of the saved document. If you specify only the file name, then it is saved in the active document's directory. The filename extension indicates the conversion that should be performed (for example, **Part1.igs** to save to IGES). If the filename extension does not uniquely indicate how the file should be formatted, use Version to specify how to save the file. For example, to save:

- A standard drawing document as a detached drawing, specify `swSaveAsDetachedDrawing` for Version.
- A detached drawing as a standard drawing, specify `swSaveAsStandardDrawing` for Version.
- A standard or detached drawing document in the same format, specify `swSaveAsCurrentVersion` for Version.

Use `ExportData` to specify which drawings sheets to save to PDF. If `ExportData` is `Nothing` or `null`, then all sheets are saved to PDF. Saving a document as PDF when the document is open as view-only is not supported.

Use Options to specify save options. You can specify additional options using [ISldWorks::SetUserPreferenceIntegerValue](#). For example:

```
' Save assembly as multibody part and save exterior faces as surface bodies
```

```
swApp.SetUserPreferenceIntegerValue swSaveAssemblyAsPartOptions, _  
    swSaveAsmAsPart_ExteriorFaces
```

```
swModelDocExt.SaveAs "H:\Assem1.SLDPRJT", swSaveAsCurrentVersion, _  
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

- or -

```
' Save all drawing sheets in active drawing document as an eDrawings file  
swApp.SetUserPreferenceIntegerValue swEdrawingsSaveAsSelectionOption, swEdrawingSaveAll  
swModelDocExt.SaveAs "H:\Grid.edrw", swSaveAsCurrentVersion, _  
    swSaveAsOptions_Silent, Nothing, nErrors, nWarnings
```

If the file is saved successfully, then the returned value is `true` and `Errors` is 0. If the save is not successful, then the returned value is `false` and `Errors` contains a bitwise OR of the error codes that were generated in saving the document. Check the masks against the `Errors` enumeration. If you do not want SOLIDWORKS to return error information, set `Errors` to `Nothing` or `null`.

Even if the file is saved successfully, there might be warnings or information that occur during the save operation in which you might be interested. `Warnings` contains a bitwise OR of the warning codes that were generated when saving the document. Check the masks against the `Warnings` enumeration. If you do not want warning information returned, set `Warnings` to `Nothing` or `null`.

Use [IModelDoc2::Save3](#) to save a file using its current name.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SaveDeFeaturedFile Method \(\)](#)
[ISldWorks::CheckpointConvertedDocument Method \(\)](#)
[ISldWorks::CloseAllDocuments Method \(\)](#)
[ISldWorks::CloseDoc Method \(\)](#)
[ISldWorks::CopyDocument Method \(\)](#)
[ISldWorks::ExitApp Method \(\)](#)
[ISldWorks::LoadFile4 Method \(\)](#)
[ISldWorks::OpenDoc7 Method \(\)](#)
[ISldWorks::QuitDoc Method \(\)](#)

Availability

SOLIDWORKS 2020 SP02, Revision Number 28.2

SOLIDWORKS API Help

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SaveDeFeaturedFile Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveDeFeaturedFile Method (IModelDocExtension)

Removes all CAD data except the outer surface from a loaded part or assembly document and saves the outer surface as a part.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveDeFeaturedFile( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.SaveDeFeaturedFile(FileName)
```

C#

```
System.bool SaveDeFeaturedFile(  
    System.string FileName  
)
```

C++/CLI

```
System.bool SaveDeFeaturedFile(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and file name for the new, de-featured part

Return Value

True if method call is successful; false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveDeFeaturedFile](#).

Example

[Save as De-Featured File \(VBA\)](#)
[Save as De-Featured File \(VB.NET\)](#)
[Save as De-Featured File \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SaveAs Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SaveDraftingStandard Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveDraftingStandard Method (IModelDocExtension)

Saves the current custom drafting standard to a file.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveDraftingStandard( _
    ByVal FileName As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim FileName As System.String
Dim value As System.Boolean

value = instance.SaveDraftingStandard(FileName)
```

C#

```
System.bool SaveDraftingStandard(
    System.string FileName
)
```

C++/CLI

```
System.bool SaveDraftingStandard(
    & System.String^ FileName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and filename for the current custom drafting standard to save to a file

Return Value

True if the current custom drafting standard is saved to a file, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveDraftingStandard](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CopyDraftingStandard Method](#)
[IModelDocExtension::DeleteDraftingStandard Method](#)
[IModelDocExtension::GetDraftingStandardNames Method](#)
[IModelDocExtension::LoadDraftingStandard Method](#)
[IModelDocExtension::RenameDraftingStandard Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SavePackAndGo Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SavePackAndGo Method (IModelDocExtension)

Saves the files designated for [Pack and Go](#) to either a folder or Zip file.

.NET Syntax

Visual Basic (Declaration)

```
Function SavePackAndGo( _
    ByVal PackAndGo As PackAndGo _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim PackAndGo As PackAndGo
Dim value As System.Object

value = instance.SavePackAndGo(PackAndGo)
```

C#

```
System.Object SavePackAndGo(
    PackAndGo PackAndGo
)
```

C++/CLI

```
System.Object^ SavePackAndGo(
    & PackAndGo^ PackAndGo
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PackAndGo
[Pack and Go](#) object

Return Value

Array of the status codes of Pack and Go as defined in [swPackAndGoSaveStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SavePackAndGo](#).

Example

[Pack and Go an Assembly \(C#\)](#)
[Pack and Go an Assembly \(VB.NET\)](#)
[Pack and Go an Assembly \(VBA\)](#)
[Add and Remove Files from Pack and Go \(C#\)](#)
[Add and Remove Files from Pack and Go \(VB.NET\)](#)
[Add and Remove Files from Pack and Go \(VBA\)](#)

Remarks

After providing Pack and Go input, call this method to execute Pack and Go and end the Pack and Go session. A subsequent call to this method will fail without again providing Pack and Go input.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SaveSelection Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveSelection Method (IModelDocExtension)

Creates a new selection set containing the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveSelection( _
    ByRef Status As System.Integer _
```

```
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Status As System.Integer  
Dim value As System.Object  
  
value = instance.SaveSelection(Status)
```

C#

```
System.object SaveSelection(  
    out System.int Status  
)
```

C++/CLI

```
System.Object^ SaveSelection(  
& [Out] System.int Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Status

1 if a selection set is created, 0 if not

Return Value

[Selection set](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveSelection](#).

Example

[Create and Delete Selection Sets \(C#\)](#)
[Create and Delete Selection Sets \(VB.NET\)](#)
[Create and Delete Selection Sets \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[ISelectionSetFolder Interface](#)
[ISelectionSetItem Interface](#)
[ISelectionSet::RemoveSelectionSet Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SaveTo3DExperience Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SaveTo3DExperience Method (IModelDocExtension)

Saves this document in [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#) using the specified save options.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveTo3DExperience( _
    ByVal Options As System.Object, _
    ByRef Errors As System.Integer, _
    ByRef Warnings As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim Options As System.Object
Dim Errors As System.Integer
Dim Warnings As System.Integer
Dim value As System.Boolean

value = instance.SaveTo3DExperience(Options, Errors, Warnings)
```

C#

```
System.bool SaveTo3DExperience(
    System.object Options,
    out System.int Errors,
    out System.int Warnings
)
```

C++/CLI

```
System.bool SaveTo3DExperience(
    & System.Object^ Options,
    & [Out] System.int Errors,
    & [Out] System.int Warnings
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options
[ISaveTo3DExperienceOptions](#) (see **Remarks**)
Errors
 Error codes
Warnings
 Warning codes

Return Value

True if the document saved successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveTo3DExperience](#).

Remarks

If the file is:

- new and the file name is specified in Options, then this method acts like **File > Save**. If any other options are specified, then this method acts like **File > Save With Options**.
- already saved to the platform and a new file name is specified in Options, then this method acts like **File > Save As New**.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2020 SP3.1, Revision Number 28.3.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

SaveViewToSolidworks Method (IModelDocExtension)

[See Also](#)



Saves the specified named view of this model to SOLIDWORKS.

.NET Syntax

Visual Basic (Declaration)

```
Function SaveViewToSolidworks( _  
    ByVal ViewName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ViewName As System.String  
Dim value As System.Boolean  
  
value = instance.SaveViewToSolidworks(ViewName)
```

C#

```
System.bool SaveViewToSolidworks(  
    System.string ViewName  
)
```

C++/CLI

```
System.bool SaveViewToSolidworks(  
& System.String^ ViewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewName
Custom view name

Return Value

True if named view successfully saved, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SaveViewToSolidworks](#).

Example

'VBA

Option Explicit

Sub main()

Dim swApp As SldWorks.SldWorks

Set swApp = Application.SldWorks

Dim swModel As ModelDoc2

Set swModel = swApp.ActiveDoc

swModel.ViewRotateplusz

swModel.GraphicsRedraw2

Dim newName As String

newName = Format(Now, "YYYYMMDD-hhmmss")

Dim bResult As Boolean

swModel.**NameView** newName

bResult = swModel.**Extension.SaveViewToSolidworks**(newName)

Debug.Print "Saved view to SOLIDWORKS as """" & newName & """" == " & bResult

End Sub

Remarks

For more information, read **SOLIDWORKS Design user-interface help > Detailing and Drawings > Drawings > Standard Drawing Views > Adding Named Views**.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments on this topic.](#)

ScaleOrCopy Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ScaleOrCopy Method (IModelDocExtension)

Scales and optionally copies the selected sketch items or annotations.

.NET Syntax

Visual Basic (Declaration)

```
Sub ScaleOrCopy( _  
    ByVal Copy As System.Boolean, _  
    ByVal NumCopies As System.Integer, _  
    ByVal BaseX As System.Double, _  
    ByVal BaseY As System.Double, _  
    ByVal BaseZ As System.Double, _  
    ByVal Scale As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Copy As System.Boolean  
Dim NumCopies As System.Integer  
Dim BaseX As System.Double  
Dim BaseY As System.Double  
Dim BaseZ As System.Double  
Dim Scale As System.Double  
  
instance.ScaleOrCopy(Copy, NumCopies, BaseX, BaseY, BaseZ, Scale)
```

C#

```
void ScaleOrCopy(  
    System.bool Copy,  
    System.int NumCopies,  
    System.double BaseX,  
    System.double BaseY,  
    System.double BaseZ,
```



Copy Code

```
        System.double Scale  
    )
```

C++/CLI

```
void ScaleOrCopy(  
    & System.bool Copy,  
    & System.int NumCopies,  
    & System.double BaseX,  
    & System.double BaseY,  
    & System.double BaseZ,  
    & System.double Scale  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Copy

True to copy the sketch items or annotations, false to not

NumCopies

Number of copies

BaseX

X coordinate of the base point

BaseY

Y coordinate of the base point

BaseZ

Z coordinate of the base point

Scale

Factor by which to scale the sketch entities or annotations

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ScaleOrCopy](#).

Example

[Rotate, Scale, and Copy Sketch \(C#\)](#)

[Rotate, Scale, and Copy Sketch \(VB.NET\)](#)

[Rotate, Scale, and Copy Sketch \(VBA\)](#)

Remarks

Before calling this method, use [IModelDocExtension::SelectByID2](#) to select the entities to scale or copy.

Using this method may break existing sketch relations, including relations that are automatically created when offsetting or converting entities. Use the [ISketchRelationManager](#) methods before and after using this method to determine whether all sketch relations remain intact.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::MoveOrCopy Method](#)

[IModelDocExtension::RotateOrCopy Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SelectAll Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SelectAll Method (IModelDocExtension)

Selects all edges in a part, all components in an assembly, or all entities (by default, sketch entities, dimensions, and annotations) in a drawing.

.NET Syntax

Visual Basic (Declaration)

```
Sub SelectAll()
```

```
Dim instance As IModelDocExtension

instance.SelectAll()
```

C#

```
void SelectAll()
```

C++/CLI

```
void SelectAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SelectAll](#).

Example

[Select All in Part, Assembly, or Drawing \(C#\)](#)
[Select All in Part, Assembly, or Drawing \(VB.NET\)](#)
[Select All in Part, Assembly, or Drawing \(VBA\)](#)

Remarks

This method behaves the same as box selecting everything in the graphics area of a part or assembly document or in a sheet of a drawing document.

To select entities different from the defaults, use selection filters. The **See Also** section contains links to methods related to selection filters.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::MultiSelect Method](#)
[IModelDocExtension::IMultiSelect Method](#)
[IModelDocExtension::SelectById2 Method](#)
[IComponent2::Select4 Method](#)
[IBody2::GetEdges Method](#)
[IBody2::IGetEdges Method](#)
[IView::ISelectEntity Method](#)
[IView::SelectEntity Method](#)
[ISldWorks::GetSelectionFilter Method](#)
[ISldWorks::IGetSelectionFilters Method](#)
[ISldWorks::IGetSelectionFilters Method](#)
[ISldWorks::IGetSelectionFiltersCount Method](#)
[ISldWorks::ISetSelectionFilters Method](#)
[ISldWorks::SetSelectionFilter Method](#)
[ISldWorks::SetSelectionFilters Method](#)
[ISldWorks::GetApplySelectionFilter Method](#)
[ISldWorks::SetApplySelectionFilter Method](#)
[ISelectData Interface](#)
[ISelectionMgr Interface](#)
[IAssemblyDoc::SelectComponentsBySize Method \(\)](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SelectById Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SelectById Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::SelectById2](#).

.NET Syntax

Visual Basic (Declaration)

```

Function SelectByID( _
    ByVal Name As System.String, _
    ByVal Type As System.String, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal Append As System.Boolean, _
    ByVal Mark As System.Integer, _
    ByVal Callout As Callout _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim Name As System.String
Dim Type As System.String
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim Append As System.Boolean
Dim Mark As System.Integer
Dim Callout As Callout
Dim value As System.Boolean

value = instance.SelectByID(Name, Type, X, Y, Z, Append, Mark, Callout)

```

C#

```

System.bool SelectByID(
    System.string Name,
    System.string Type,
    System.double X,
    System.double Y,
    System.double Z,
    System.bool Append,
    System.int Mark,
    Callout Callout
)

```

C++/CLI

```

System.bool SelectByID(
    & System.String^ Name,
    & System.String^ Type,
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.bool Append,
    & System.int Mark,
    & Callout^ Callout
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Type

X

Y

Z

Append

Mark

Callout

See [ModelDocExtension::SelectByID](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SelectByID2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SelectByID2 Method (IModelDocExtension)

Selects the specified entity.

VB.NET Syntax

Visual Basic (Declaration)

```
Function SelectByID2( _  
    ByVal Name As System.String, _  
    ByVal Type As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double, _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer, _  
    ByVal Callout As Callout, _  
    ByVal SelectOption As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim Type As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim Append As System.Boolean  
Dim Mark As System.Integer  
Dim Callout As Callout  
Dim SelectOption As System.Integer  
Dim value As System.Boolean  
  
value = instance.SelectByID2(Name, Type, X, Y, Z, Append, Mark, Callout, SelectOption)
```

C#

```
System.bool SelectByID2(  
    System.string Name,  
    System.string Type,  
    System.double X,  
    System.double Y,  
    System.double Z,  
    System.bool Append,  
    System.int Mark,  
    Callout Callout,  
    System.int SelectOption  
)
```

C++/CLI

```
System.bool SelectByID2(  
& System.String^ Name,  
& System.String^ Type,  
& System.double X,  
& System.double Y,  
& System.double Z,
```

```
& System.bool Append,
& System.int Mark,
& Callout^ Callout,
& System.int SelectOption
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of object to select or an empty string

Type

Type of object (uppercase) as defined in [swSelectType_e](#) or an empty string

X

X selection location or 0

Y

Y selection location or 0

Z

Z selection location or 0

Append

If...	An, if entity is...	Then...
True	Not already selected	Entity is appended to the current selection list
	Already selected	Entity is removed from the current selection list
False	Not already selected	Current selection is cleared and then the entity is put on the list
	Already selected	Current selection list remains the same

Mark

Value that you want to use as a mark; this value is used by other functions that require ordered selection (see **Remarks**)

Callout

Pointer to the associated [callout](#)

SelectOption

Selection option as defined in [swSelectOption_e](#) (see **Remarks**)

Return Value

True if item was successfully selected, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SelectByID2](#).

Example

[Access Assembly \(C++\)](#)
[Add Component and Mate \(C++\)](#)
[Create Cylinder \(C++\)](#)
[Traverse Bodies \(C++\)](#)
[Add Distance Mates \(VBA\)](#)
[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Get Selected Object \(VBA\)](#)
[Insert Wrap Feature \(VBA\)](#)
[Select Bodies \(VBA\)](#)
[Connect to SOLIDWORKS Session \(C#\)](#)
[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)
[Select Multiple Sketch Segments for Sweep Path \(C#\)](#)
[Select Multiple Sketch Segments for Sweep Path \(VB.NET\)](#)
[Select Multiple Sketch Segments for Sweep Path \(VBA\)](#)
[Select Multiple Splines for Loft Guide Curves \(C#\)](#)
[Select Multiple Splines for Loft Guide Curves \(VB.NET\)](#)
[Select Multiple Splines for Loft Guide Curves \(VBA\)](#)

Remarks

Use this method instead of using the selection methods on the following objects:

- [IAnnotation](#)
- [IComponent2](#)
- [IFeature](#)
- [IFeatureManager](#)
- [ISketchHatch](#)
- [ISketchPoint](#)
- [ISketchSegment](#)

- [ISketchSpline](#)

The previously listed objects' selection methods do not work well when a PropertyManager page is open or a command is running. This method, `IModelDocExtension::SelectByID2`, handles selection correctly regardless if a command is running.

If your application already has an object handle (for example, `IFace2`), use the appropriate Select method to select the item directly using its handle.

To filter the objects selected by this method, set the Type parameter. If no filtering is required, then pass an empty string for this parameter.

For example, to select an object of type `swSelDIMENSIONS`, use the string that appears in the comment column, "DIMENSION". The object Type might change based on your current state. For example, to select a sketch point that was created in the active sketch, specify Type as "SKETCHPOINT". However, if you do not have an active sketch, or if the point was created in a sketch other than the active sketch, or the point is the origin point, then specify Type as "EXTSKETCHPOINT".

If Type is specified, then this method returns false if it cannot find the matching object type.

The Name parameter is not intended for selection of faces, edges, and so on. This is a case-sensitive string for objects that are automatically named by SOLIDWORKS during entity creation, such as dimensions, drawing views, and so on. This method tries to find and select an object whose name matches the Name parameter; however, the match needs to be exact for this method to return true.

For example:

- If a string is passed that matches an object name but whose case does not match exactly, then this method could return false. For selection of dimensions, the Name parameter must be fully qualified.
- Specify "D1@Sketch2@Part1.SLDPRJT" rather than "D1@Sketch2"; otherwise, this method could return false. If you do not know the object name, or if it is an item that is not automatically named by SOLIDWORKS, you can pass an empty string.

If you are using the Name parameter, then specify the x, y, and z coordinates in terms of the context where the item was created. For example, you want to select a sketch point using its name (for example, "Point1") in the Name parameter, then specify X, Y, and Z in terms of the sketch where the point was created. Even if the sketch is not active, specify the X, Y, and Z values in terms of sketch space when you use the Name parameter. In certain situations, you can also pass in the x, y, and z coordinates as (0,0,0). For example, to select a feature shown in the FeatureManager design tree, you do not need to specify x, y, and z coordinates. However, to select objects such as faces or when you need a point location picked, you must specify the x, y, and z coordinates. If you are not using the Name parameter as a filter, then specify the x, y, and coordinates in terms of model space. The coordinates are used when the object name is not provided, and the coordinates are dependent on the view state.

If you do not know the object name or the object type, pass empty strings for the Name and Type parameters. The selection routine makes the best attempt to select the correct object.

To get [IFace2](#), [IEdge](#) or [IVertex](#) objects by name, use [IPartDoc::GetEntityByName](#) or [IPartDoc::IGetEntityByName](#).

For `SelectOption`, specify `swSelectOption_e.swSelectOptionDefault` to indicate that the Shift key is not used during selection or specify `swSelectOption_e.swSelectOptionExtensive` to indicate that the Shift key is used during selection.

The [IModelDoc2](#) object used to call this method must be an open and visible document. For example, you cannot use the `IModelDoc2` object returned from an assembly component ([IComponent2::GetModelDoc](#) or [IComponent2::IGetModelDoc](#)) unless that SOLIDWORKS component part or subassembly is an open and visible document. In this case, you can select the item using the fully qualified name (for example, "Plane4@Part1-1@Assem1").

When selecting `IFace2` objects, this method uses the specified point as input to the normal user-interface selection routines to use the speed of ray tracing. As a result, if the view changes from the original recorded size or orientation, then the same `IFace2` might not be selected next time. If your application has a pointer to the `IFace2` object to be selected, then you can call the [IEntity::Select4](#) method directly. Otherwise, you can call [IModelDocExtension::SelectByRay](#). Calling either of these methods allows for tighter control over the starting point and the direction in which to search. `IModelDocExtension::SelectByRay` is recorded when `IModelDocExtension::SelectByID2` relies on the selection coordinates and prone to failure if the model view is altered.

Use:

- Mark of 4 when selecting multiple edges or sketch segments and grouping them into one object for the path for a sweep feature. See the **Select Multiple Sketch Segments for Sweep Path** examples.
- Mark of 2 when selecting multiple edges, sketch segments, or curves and grouping them into one object for the guide curves for a loft feature. See the **Select Multiple Splines for Loft Guide Curves** examples.
- [ISelectionMgr::AddSelectionListObject](#) and [ISelectionMgr::AddSelectionListObjects](#) to add objects to a selection list without preselecting the objects in the user interface.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::MultiSelect Method](#)
[IModelDocExtension::IMultiSelect Method](#)
[ISelectData Interface](#)
[ISelectionMgr Interface](#)
[IFeature::GetNameForSelection Method](#)
[IModelDocExtension::SelectAll Method](#)
[IAssemblyDoc::SelectComponentsBySize Method](#)
[ISelectionMgr::GetSelectByIdSpecification Method](#)
[ISelectionMgr::GetSelectionSpecification Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SelectByRay Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : [SelectByRay Method \(IModelDocExtension\)](#)

Selects the first entity of the specified type that is intersected by a ray that starts at the specified point and runs parallel to the specified direction vector within the specified radius.

.NET Syntax

Visual Basic (Declaration)

```
Function SelectByRay( _  
    ByVal WorldX As System.Double, _  
    ByVal WorldY As System.Double, _  
    ByVal WorldZ As System.Double, _  
    ByVal RayVecX As System.Double, _  
    ByVal RayVecY As System.Double, _  
    ByVal RayVecZ As System.Double, _  
    ByVal RayRadius As System.Double, _  
    ByVal TypeWanted As System.Integer, _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer, _  
    ByVal Option As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim WorldX As System.Double  
Dim WorldY As System.Double  
Dim WorldZ As System.Double  
Dim RayVecX As System.Double  
Dim RayVecY As System.Double  
Dim RayVecZ As System.Double  
Dim RayRadius As System.Double  
Dim TypeWanted As System.Integer  
Dim Append As System.Boolean  
Dim Mark As System.Integer  
Dim Option As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SelectByRay(WorldX, WorldY, WorldZ, RayVecX, RayVecY, RayVecZ, RayRadius,  
TypeWanted, Append, Mark, Option)
```

C#

```
System.bool SelectByRay(  
    System.double WorldX,  
    System.double WorldY,  
    System.double WorldZ,  
    System.double RayVecX,  
    System.double RayVecY,  
    System.double RayVecZ,  
    System.double RayRadius,  
    System.int TypeWanted,  
    System.bool Append,  
    System.int Mark,  
    System.int Option  
)
```

C++/CLI

```
System.bool SelectByRay(  
& System.double WorldX,  
& System.double WorldY,  
& System.double WorldZ,  
& System.double RayVecX,  
& System.double RayVecY,  
& System.double RayVecZ,  
& System.double RayRadius,  
& System.int TypeWanted,  
& System.bool Append,  
& System.int Mark,
```

& System.int *Option*
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WorldX
x coordinate of ray start point
WorldY
y coordinate of ray start point
WorldZ
z coordinate of ray start point
RayVecX
x coordinate of ray direction vector
RayVecY
y coordinate of ray direction vector
RayVecZ
z coordinate of ray direction vector
RayRadius
Radius <= 2.9 m
TypeWanted
Type of entities to select as defined in [swSelectType_e](#)
Append

If...	And if entity is...	Then...
True	Not already selected	Entity is appended to the current selection list
	Already selected	Entity is removed from the current selection list
False	Not already selected	Current selection is cleared and then the entity is put on the list
	Already selected	Current selection list remains the same

Mark
Value to use as a mark; this value is used by other functions that require ordered selection
Option
Selection option as defined in [swSelectOption_e](#) (see **Remarks**)

Return Value

True if the entity is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SelectByRay](#).

Example

[Select Face Using Intersecting Ray \(C#\)](#)
[Select Face Using Intersecting Ray \(VB.NET\)](#)
[Select Face Using Intersecting Ray \(VBA\)](#)

Remarks

This method:

- Defines a cylindrical region of infinite length that starts at the specified point and runs parallel to the specified direction vector within the specified radius.
- Only selects these entities: faces, edges, and vertices.
- Does not support the selection of sketch entities, axes, center marks, center lines, section lines, etc.

Only a single entity is selected within the radius regardless if multiple entities existing within that radius. If selecting faces, then the RayRadius parameter is ignored and a cylinder of an infinitely small radius is used.

Use model space view to determine the selection vector in a drawing.

For the Option parameter, specify swSelectOption_e.swSelectOptionDefault to indicate that the Shift key is not used during selection or swSelectOption_e.swSelectOptionExtensive to indicate that the Shift key is used during selection.

NOTES:

- This method is recorded when [IModelDocExtension::SelectById2](#) relies on the selection coordinates and prone to failure if the model view is altered.
- The difference between this method and the now obsolete [IModelDoc2::SelectByRay](#) method is that if you call IModelDocExtension::SelectByRay in a macro or application to select an entity by an intersecting ray and then rerun that macro or application, the original entity is successfully selected regardless of the viewing transform.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::GetRayIntersectionsPoints Method \(\)](#)
[IModelDoc2::GetRayIntersectionsTopology Method \(\)](#)
[IModelDoc2::MultiSelectByRay Method \(\)](#)
[IModelDoc2::RayIntersections Method \(\)](#)

Availability

SetAdvancedSpotLightProperties Method (IModelDocExtension)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetAdvancedSpotLightProperties Method (IModelDocExtension)

Sets the attenuation-related, advanced properties for the specified SOLIDWORKS spot light in this model.

 .NET Syntax

Visual Basic (Declaration)

```

Sub SetAdvancedSpotLightProperties( _
    ByVal Name As System.String, _
    ByVal Exponent As System.Double, _
    ByVal AttenuationConst As System.Double, _
    ByVal AttenuationLinear As System.Double, _
    ByVal AttenuationQuad As System.Double _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim Name As System.String
Dim Exponent As System.Double
Dim AttenuationConst As System.Double
Dim AttenuationLinear As System.Double
Dim AttenuationQuad As System.Double

instance.SetAdvancedSpotLightProperties(Name, Exponent, AttenuationConst, AttenuationLinear,
AttenuationQuad)

```

C#

```

void SetAdvancedSpotLightProperties(
    System.string Name,
    System.double Exponent,
    System.double AttenuationConst,
    System.double AttenuationLinear,
    System.double AttenuationQuad
)

```

C++/CLI

```

void SetAdvancedSpotLightProperties(
    & System.String^ Name,
    & System.double Exponent,
    & System.double AttenuationConst,
    & System.double AttenuationLinear,
    & System.double AttenuationQuad
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).**Parameters**

Name
SOLIDWORKS light source name

Exponent
Attenuation exponent

AttenuationConst
Constant attenuation factor

AttenuationLinear
Linear attenuation factor

AttenuationQuad
Quadratic attenuation factor

 Visual Basic for Applications (VBA) SyntaxSee [ModelDocExtension::SetAdvancedSpotLightProperties](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetApiUndoObject Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetApiUndoObject Method (IModelDocExtension)

Implements an undo object for an add-in application.

.NET Syntax

Visual Basic (Declaration)

```
Function SetApiUndoObject( _  
    ByVal PHandler As System.Object, _  
    ByVal DisplayName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PHandler As System.Object  
Dim DisplayName As System.String  
Dim value As System.Boolean
```

```
value = instance.SetApiUndoObject(PHandler, DisplayName)
```

C#

```
System.bool SetApiUndoObject(  
    System.object PHandler,  
    System.string DisplayName  
)
```

C++/CLI

```
System.bool SetApiUndoObject(  
& System.Object^ PHandler,  
& System.String^ DisplayName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PHandler

[Undo object](#)

DisplayName

Name to display in the SOLIDWORKS undo list

Return Value

True if the undo object is implemented, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetApiUndoObject](#).

Example

[Automate Add-in's Undo Commands \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0